CIRCUIT main board — schematic status

Drawn (draft, from KiCad library pinouts): 2 USB-C input; 3 Haptics, buttons, debug; 4 Main MCU

Not drawn yet:
- Charger, battery, fuel gauge; 3V3 buck-boost, load switches, backlight driver;
  secure element; NFC; BLE module — these need pinouts from datasheets (no library symbols)
- Display and touch connectors — need the chosen panel's FPC pinout

Signals between sheets use global labels. No routing before this schematic is complete and reviewed.

USB-C input, CC, ESD

File: usb.kicad_sch

Haptics, buttons, debug

File: hmi.kicad_sch

Main MCU STM32U5G9

File: mcu.kicad_sch

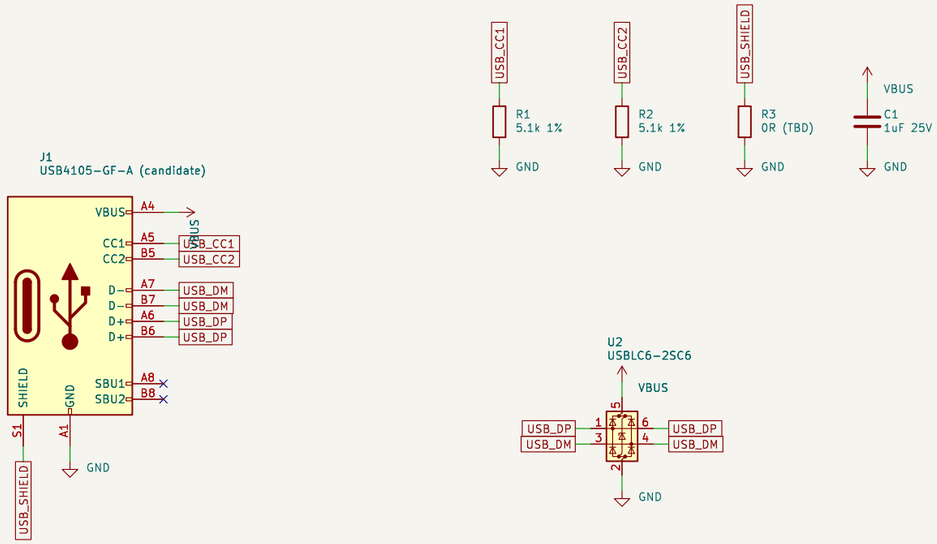
DRAFT — generated by hardware/scripts/gen_schematic.py — not reviewed

Sheet: /
File: circuit.kicad_sch

Title: Circuit main board

Size: A3	Date:	**Rev: A0-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

USB-C INPUT (5 V sink, USB 2.0 device, no Power Delivery)
- R1/R2: 5.1k Rd on each CC pin separately (USB Type-C sink requirement).
- USB_CC1/USB_CC2 go to MCU ADC inputs to read source current (0.5/1.5/3 A).
- U2 clamps D+/D- and VBUS. Both pins of each I/O pair are one node: route through the part.
- C1 kept small: a sink may present at most 10 uF on VBUS.
TODO before review: 20 V-tolerant CC protection (TPD4S311-class, no library symbol yet);
VBUS over-voltage is handled by the charger input rating - confirm from charger datasheet;
shield-to-ground scheme (R3) to be decided with the enclosure ESD path.



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Sheet: /USB-C input, CC, ESD/
File: usb.kicad_sch

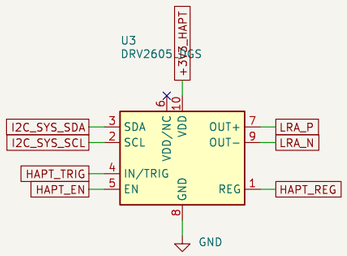
Title: USB-C input, CC, ESD

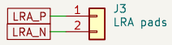
Size: A3	Date:	**Rev: A0-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/4

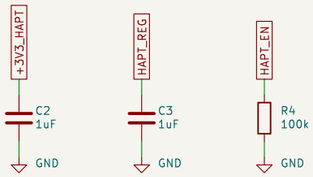
HAPTICS: DRV2605L drives an 8 mm LRA from the switched rail +3V3_HAPT.
R4 holds EN low while the MCU is in reset. Pin 6 (VDD/NC) left open — CHECK datasheet.
C2/C3 values (1 uF) are defaults — CHECK datasheet.

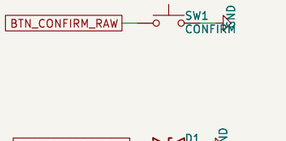
BUTTONS: active low. Pull-up, series resistor, RC debounce and ESD clamp per button.
The MCU reads BTN_CONFIRM_N directly in the secure world; it is the only signing approval input.
BTN_POWER_RAW also goes to the charger push-button input (power sheet, not drawn yet).
Switch and TVS part numbers TBD (candidates: Panasonic EVQ-P7 side-push; 0402 ESD diode).

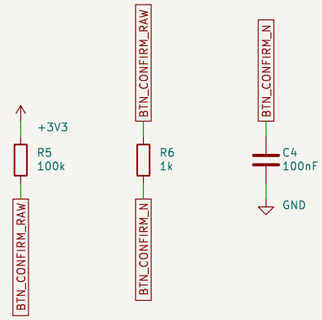
DEBUG: Tag-Connect TC2030-NL pads,
no connector fitted. Internal only.
Production: debug locked in the MCU;
pads not reachable with the case closed.

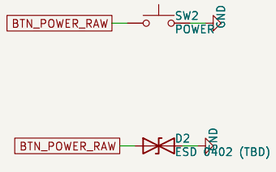


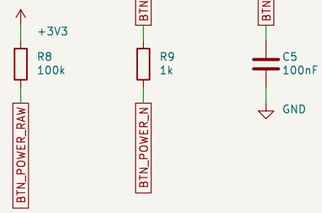


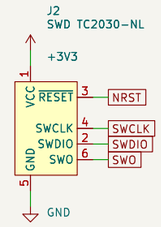












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Sheet: /Haptics, buttons, debug/
File: hmi.kicad_sch

Title: Haptics, buttons, debug

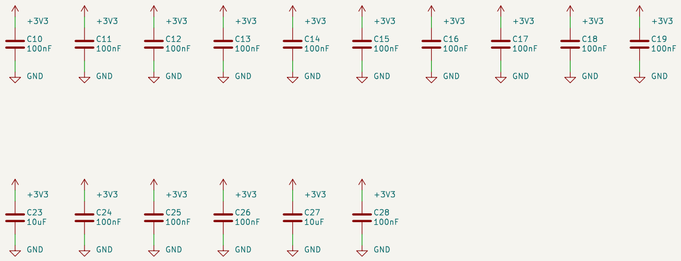
Size: A3	Date:	**Rev: A0-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/4

MAIN MCU: STM32U5G9NJH6Q, TFBGA-216 13x13 mm, 0.8 mm pitch. Pin functions are taken from the KiCad library's alternate-function data. Every ball not listed in PINMAP is left unconnected (spare GPIO).

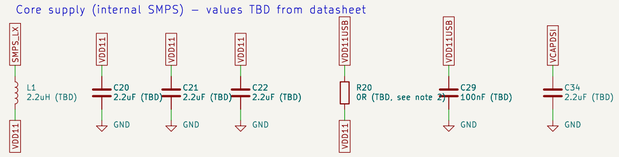
MUST BE CONFIRMED AGAINST THE DATASHEET AND AN5373 BEFORE REVIEW (st.com was not reachable):
1. SMPS network: L1 value and the capacitor on each VDD11 ball (C20-C22).
2. VDD11USB: whether it is tied to VDD11 (R20 fitted) or only decoupled (R20 not fitted).
3. DSI unused: required connection of VDDDSI, VDD11DSI/VCAPDSI and the DSI balls.
4. VBAT: tied to +3V3 through R21 for now; becomes the coin-cell backup rail if tamper-in-storage is adopted.
5. HSE 16 MHz (as on ST's dev kit) is assumed valid for the USB HS PHY; crystal load capacitors C30-C33 are placeholders.
6. Decoupling count and values (C10-C19, C23-C29) are defaults.

Nets ending in another sheet that is not drawn yet (display, touch, secure element, BLE, NFC, charger) are single-ended for now.

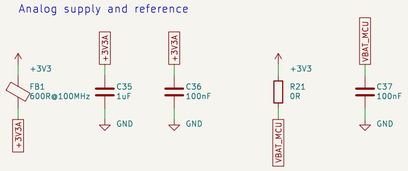
Decoupling: C10-C19 at VDD balls; C24/C25 VDDIO2; C26 VDDUSB; C27 VDDSMPS; C28 VDDDSI



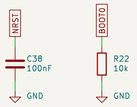
Core supply (internal SMPS) - values TBD from datasheet



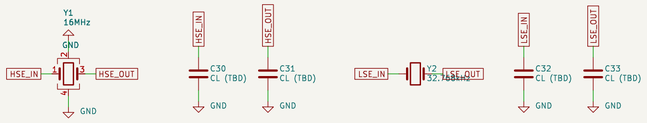
Analog supply and reference

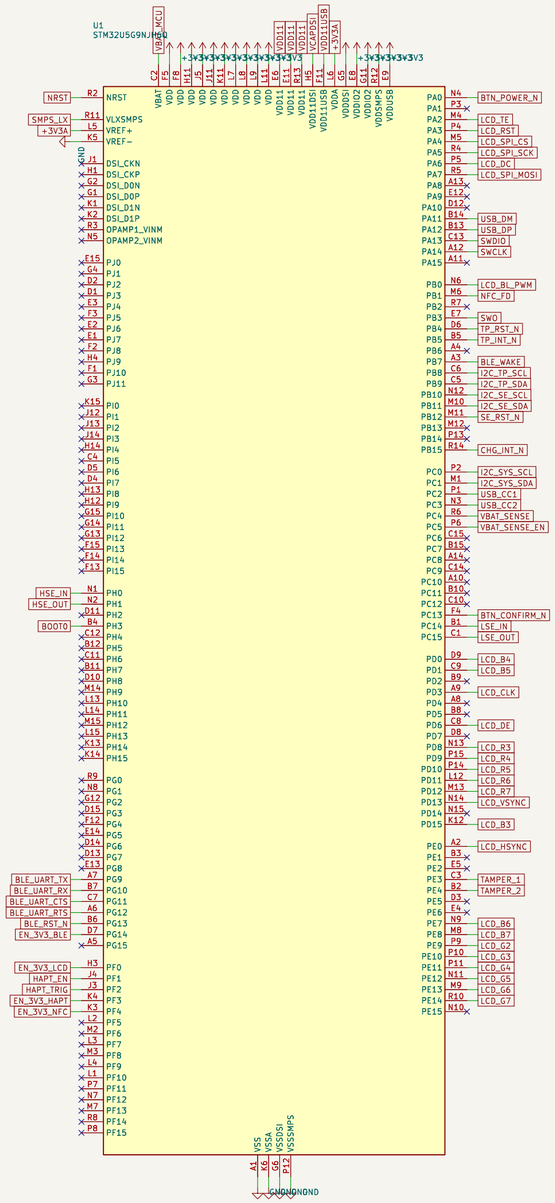


Reset and boot

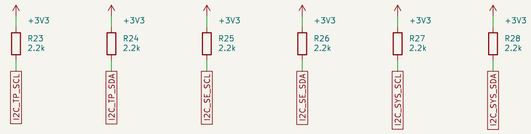


Clocks: 16 MHz HSE, 32.768 kHz LSE (load capacitors depend on the crystals chosen)





Shared pull-ups for the three I2C buses (values default)



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Sheet: /Main MCU STM32U5G9/
File: mcu.kicad_sch

Title: Main MCU STM32U5G9

Size: A2 | Date: | **Rev: A0-draft**
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